Lecture 05: Probability and Statistical Inference

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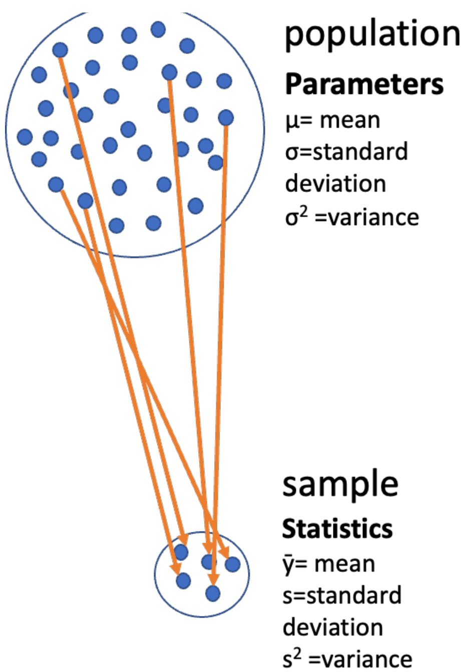
Lecture 4: Review

- Introduction to histograms or frequency distributions
- Probability Distribution Functions (PDF)
- Descriptive Statistics
  - Center - mean, median, mode
  - Spread - range, variance, standard deviation
- Tests of means using T-Tests
  - one sample
  - two sample



Lecture 4: Review - Statistical Concepts

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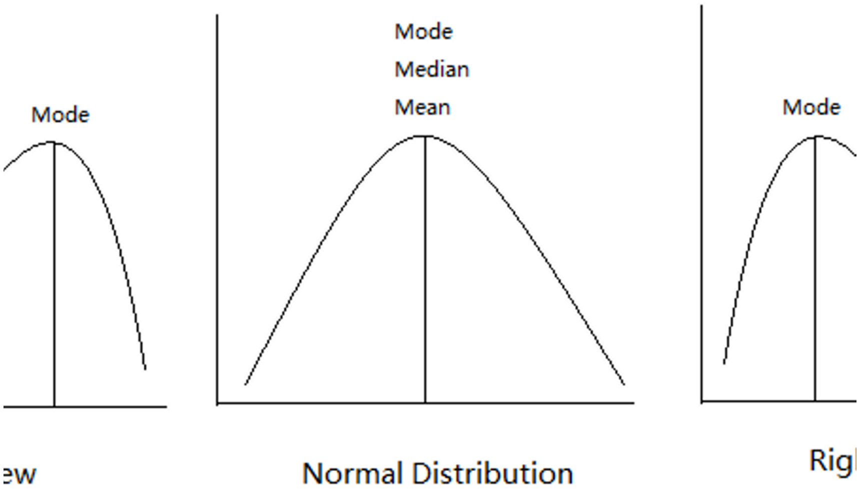
  - Center - mean, median, mode
  - Spread - range, variance, standard deviation
- Tests of means using T-Tests
  - one sample
  - two sample



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Lecture 4: Review - Summary Statistics

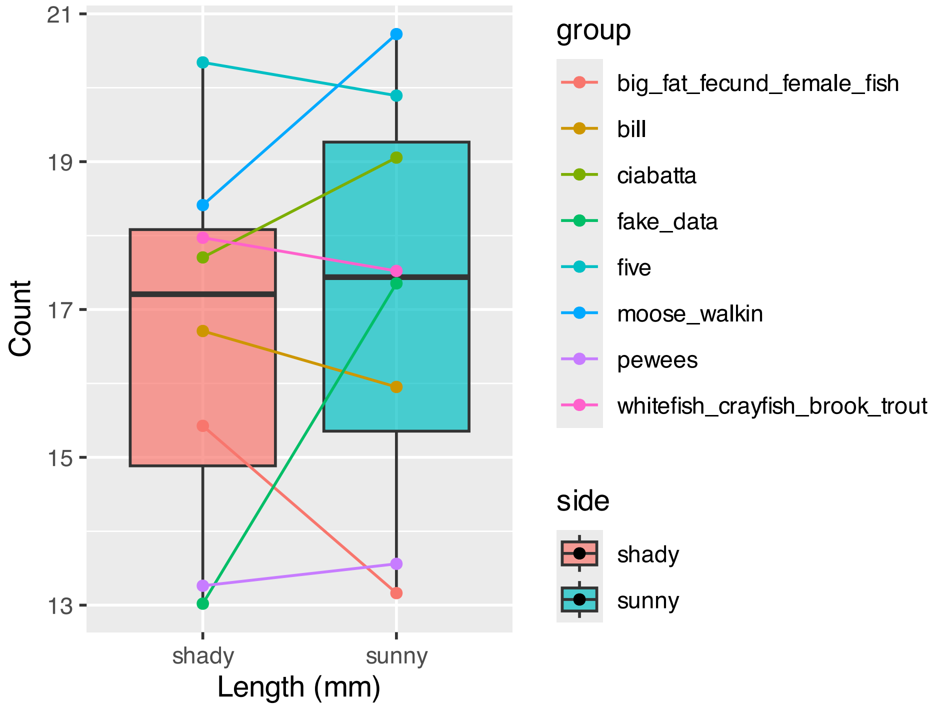
- Introduction to histograms or frequency distributions
- Probability Distribution Functions (PDF)
- Descriptive Statistics
  - Center - mean, median, mode
  - Spread - range, variance, standard deviation
- Tests of means using T-Tests
  - one sample - is the sample mean different from a hypothesized mean?
  - two sample - are the sample means from two samples the same or different?
- for two sample T tests the df = n1+ n2 −2 = 8+8-2=14

```
# A tibble: 2 × 5
  side  mean_length sd_length se_length count
  <chr>       <dbl>     <dbl>     <dbl> <int>
1 shady        17.6      2.51     0.886     8
2 sunny        16.2      2.64     0.934     8
```

Lecture 5: Probability and Statistical Inference

The goals for today

- Statistical inference fundamentals
- Hypothesis testing principles
- T Distributions
- One sample T Tests
- Two sample T Test



Pine needles

Lecture 5: Probability and Statistical Inference

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- T Distributions
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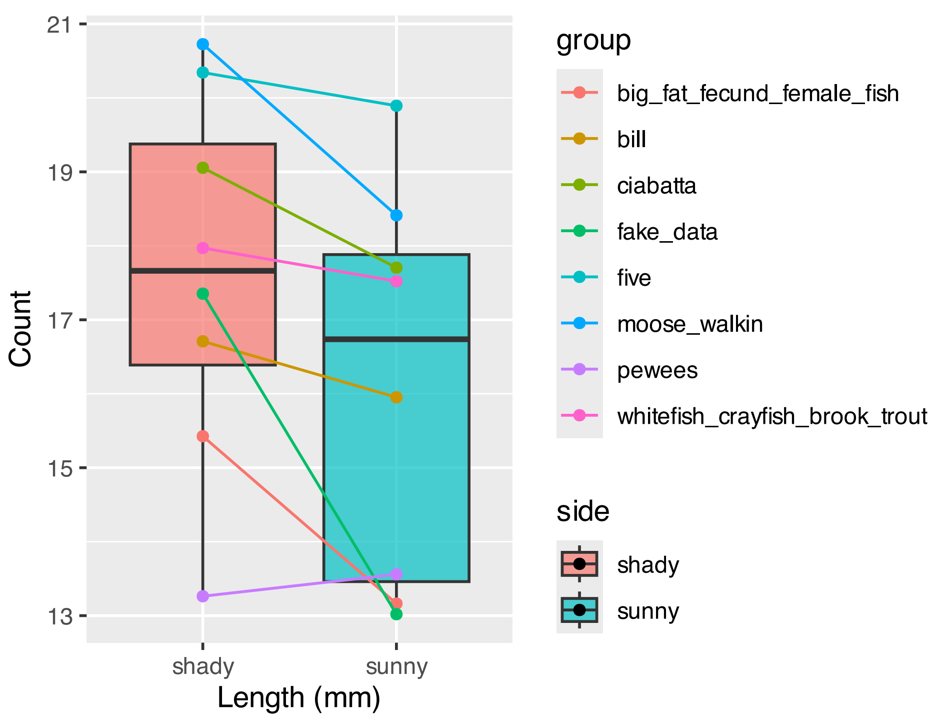
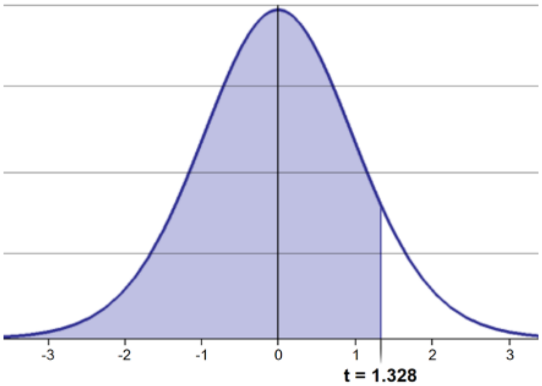


Pine needles switched

Lecture 5: One-tailed Questions

One-tailed questions: area of distribution left or (right) of a certain value for a one sample test

- n=8 (df=7) - 95% of the observations found left
- t= 1.761 (5% are outside)



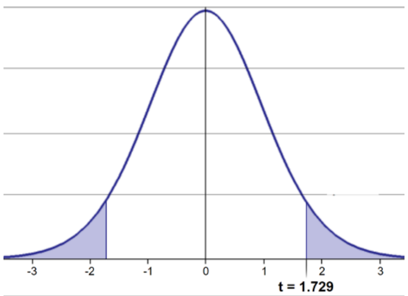
cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

xxxx

Lecture 5: Two-tailed Questions

Two-tailed questions refer to area between certain values

- n= 20 (df=19), 90% of the observations are between
- t=-1.729 and t=1.729 (10% are outside)



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	Confidence Level										

xxxxx

Lecture 5: Calculating CI Example

Let's calculate CIs again:

Use two-sided test

- CI $= \bar{y} \pm t \cdot \frac{s}{\sqrt{n}}$
- 95% CI Sample A: = 17.6 ± 2.365 * (2.51/(8^0.5)) = +/- 2.098746
- The 95% CI is between 15.50 and 19.70
- "The 95% CI for the population mean from sample A is 17.6 ± 2.1

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	Confidence Level										

Lecture 5: Applications of t-distribution

So:

- Can assess confidence that population mean is within a certain range
- Can use t distribution to ask questions like:
  - "What is probability of getting sample with mean = $\bar{y}$ from population with mean = μ?" (1 sample t-test)
  - "What is the probability that two samples came from same population?" (2 sample t-test)

Lecture 5: Single Sample T-Test

We want to test if the mean fish length in I3 differs from 240mm.

Activity: Define hypotheses and identify assumptions

H_0: μ = 15 (The mean needle length on shade side is 15mm)

H_1: $\mu \neq$ 15 (The mean needle length on shade side is not 240mm)

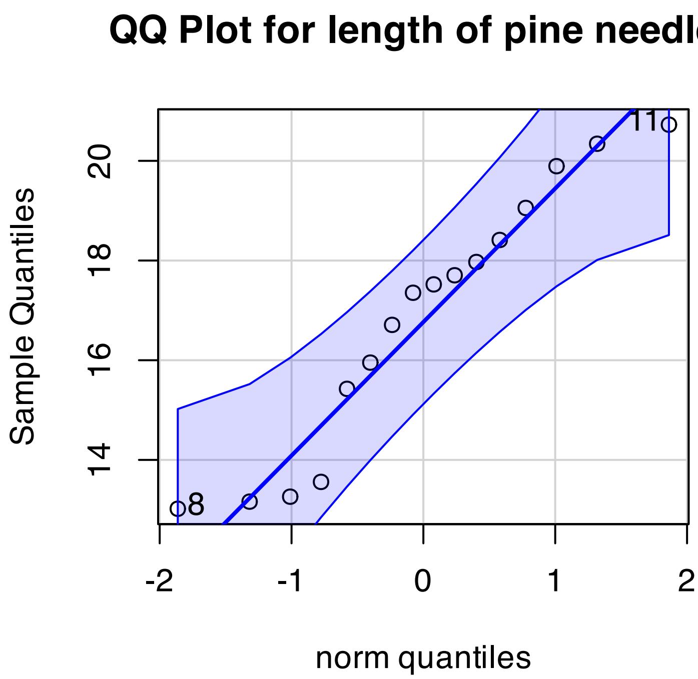
Assumptions for t-test:

1. Data is normally distributed
2. Observations are independent
3. No significant outliers

Assumptions in R - qqplots from car

```
# Filter for just windward side needles

# YOUR TASK: Test normality of windward pine needle lengths
# QQ Plot
qqPlot(ps_df$length_mm,
```

```
       main = "QQ Plot for length of pine needles",
       ylab = "Sample Quantiles")
```




```
[1]  8 11
```

Statistical Test of Normality

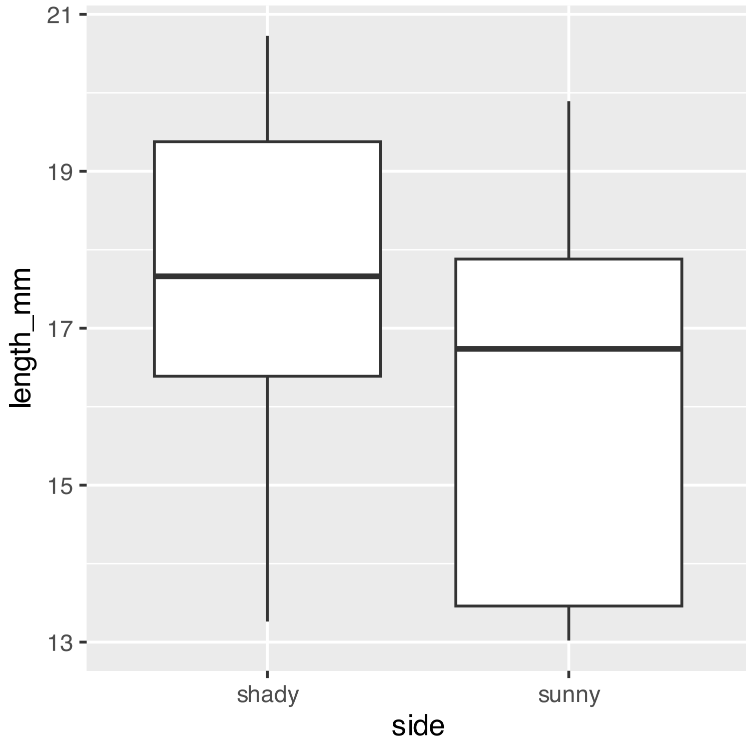
Shapiro-Wilk test

```
# Shapiro-Wilk test
shapiro.test(ps_df$length_mm)
```

```

	Shapiro-Wilk normality test

data:  ps_df$length_mm
W = 0.92754, p-value = 0.2228
```

Checking for Outliers

```
# Check for outliers using boxplot
# YOUR CODE HERE
ps_df %>% ggplot(aes(side, length_mm))+geom_boxplot()
```



Practice Exercise 1: One-Sample t-Test

Practice Exercise 1: One-Sample t-Test

Let's perform a one-sample t-test to determine if the mean fish length in I3 Lake differs from 15 mm:

```
# what is the mean

ps_shade_mean <- mean(ps_shady_df$length_mm, na.rm = TRUE)
cat("Mean:", round(ps_shade_mean, 1), "mm\n")
```

```
Mean: 17.6 mm
```

```
# Perform a one-sample t-test
t_test_result <- t.test(ps_shady_df$length_mm, mu = 15)

# View the test results
t_test_result
```

```

	One Sample t-test

data:  ps_shady_df$length_mm
t = 2.9414, df = 7, p-value = 0.02167
alternative hypothesis: true mean is not equal to 15
95 percent confidence interval:
 15.51092 19.70030
sample estimates:
mean of x 
 17.60561 
```

Interpret this test result by answering these questions:

1. What was the null hypothesis?
2. What was the alternative hypothesis?
3. What does the p-value tell us?
4. Should we reject or fail to reject the null hypothesis at $\alpha = 0.05$?
5. What is the practical interpretation of this result for fish biologists?

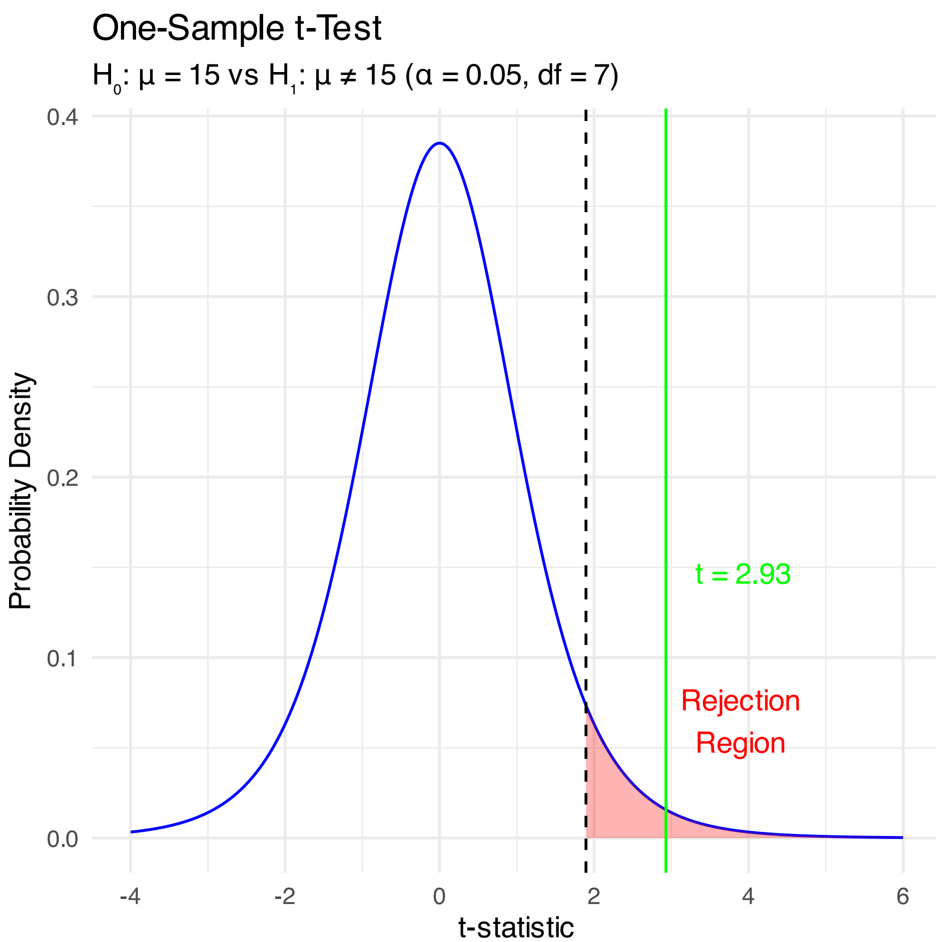
Lecture 5: Hypothesis Testing Framework

Hypothesis testing is a systematic way to evaluate research questions using data.

Key components:

1. **Null hypothesis (Ho)**: Typically assumes "no effect" or "no difference"
2. **Alternative hypothesis (Ha)**: The claim we're trying to support
3. **Statistical test**: Method for evaluating evidence against H_0
4. **P-value**: Probability of observing our results (or more extreme) if H_0 is true
5. **Significance level (α)**: Threshold for rejecting H_0, typically 0.05

Decision rule: Reject Ho if p-value less than α or shorthand $p < 0.05$



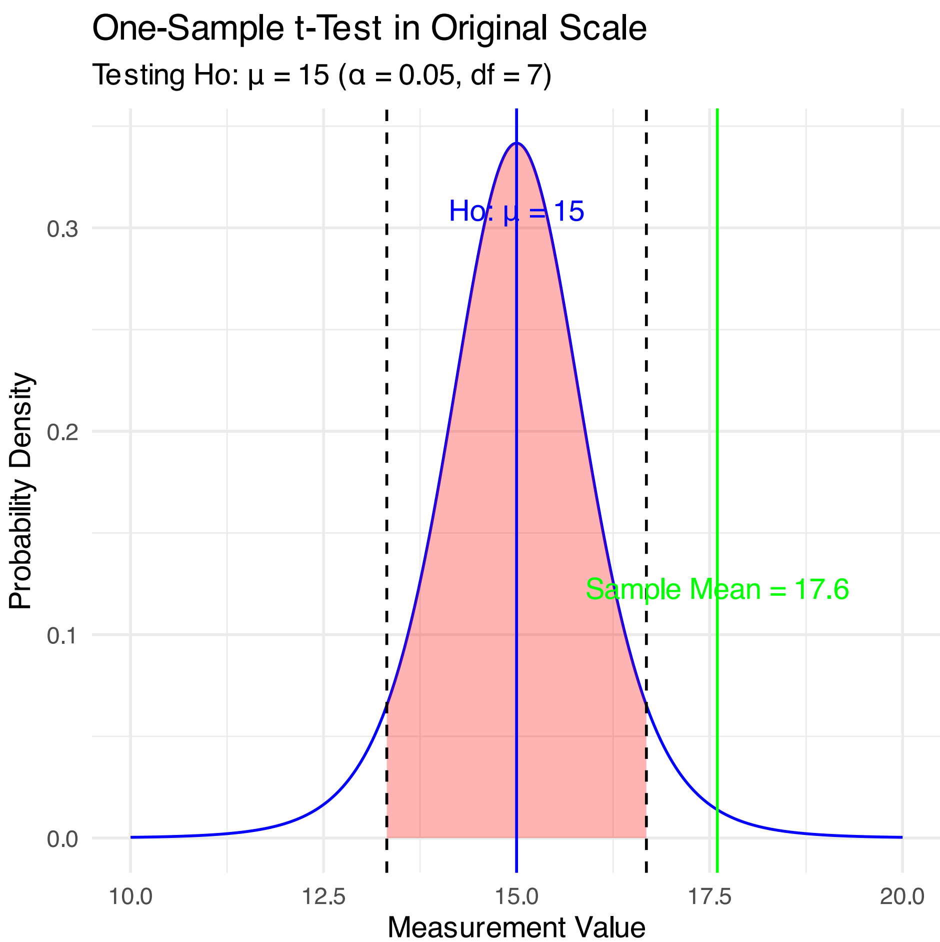
Lecture 5: Hypothesis Testing

Hypothesis testing is a systematic way to evaluate research questions using data.

Key components:

1. **Null hypothesis (H_0)**: Typically assumes "no effect" or "no difference"
2. **Alternative hypothesis (H_a)**: The claim we're trying to support
3. **Statistical test**: Method for evaluating evidence against H_0
4. **P-value**: Probability of observing our results (or more extreme) if H_0 is true
5. **Significance level (α)**: Threshold for rejecting H_0, typically 0.05

Decision rule: Reject H_0 if p-value $< \alpha$



Lecture 5: Interpreting One-Sample T-Test Results

Activity: Interpret the t-test results

- What does the p-value tell us?
- Should we reject or fail to reject the null hypothesis?

How to report this result in a scientific paper:

"A one-sample t-test at α=0.05 showed that the mean needle length (… mm, SD = …) [was/was not] significantly different from the expected 15 mm, t(…) = …, p = …"

Lecture 5: Two Sample T-Tests Introduction

For example

- what is probability that population X is the same as population Y?

How would you assess this question using what we learned?

This is what we will do with the fish length again…

xxxxx  xxx

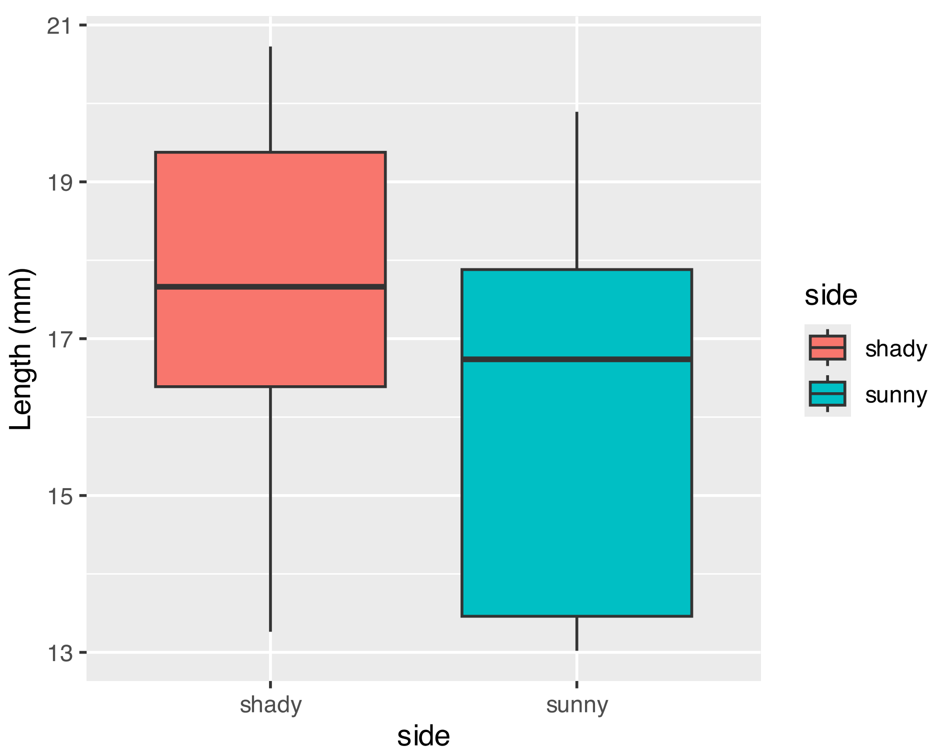
Lecture 5: Comparing Two Samples

For example

- what is probability that population X is the same as population Y?

How would you assess this question using what we learned?

```
# Now create a boxplot to visualize the difference in fish lengths between these lakes:


# Create a boxplot comparing the two lakes
shady_sunny_plot <- ps_df %>%
  ggplot(aes(x = side, y = length_mm, fill = side)) +
  geom_boxplot() +
  labs(
       x = "side",
       y = "Length (mm)",
       fill = "side")
shady_sunny_plot
```



```
# Based on the t-test results and the boxplot
#
# what can you conclude about the fish populations in these two lakes?
```

Practice Exercise 2: Formulating Hypotheses

Practice Exercise 2: Formulating Hypotheses

For the following research questions about fish lengths write the null and alternative hypotheses:

1. Are needle lengths ion shady and sunny sides different?

What are the hypotheses?

Ho =

Ha =

Lecture 5: Two-Sample T-Test Framework

Now, let's compare needles lengths from the two sides

Question: **Is there a significant difference in neele length between the sides?**

This requires a two-sample t-test.

Two-sample t-test compares means from two independent groups.

$$t = \frac{\bar{x_1} - \bar{x_2}}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where:

- $\bar{x}_1$ and $\bar{x}_2$: These represent the sample means of the two groups you're comparing
- ${s^2}_p$: This is the pooled variance, calculated as: ${s^2}_p = [(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2] / (n_1 + n_2 - 2)$, where s_1^2 and s_2^2 are the sample variances of the two groups.
- **n_1 and n_2:** These are the sample sizes of the two groups.
- **$\sqrt{(1/n_1 + 1/n_2)}$:** This represents the pooled standard error.

$$t = \frac{SIGNAL}{NOISE}$$

Practice Exercise 3: Summary Statistics

Practice Exercise 3: **Calculate summary statistics grouped by lake**

Before conducting the test, we need to understand the data for each group.

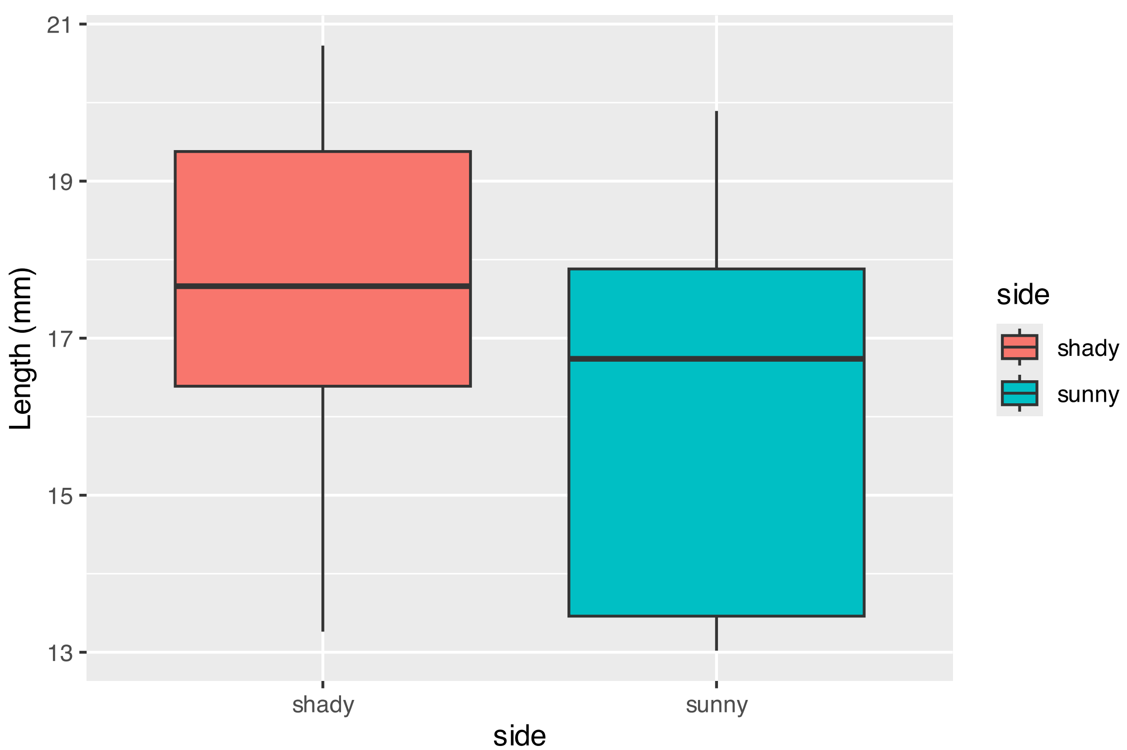
1. You need this and the graph to see what is going on ….

```
group_summary <- ps_df %>%
  group_by(side) %>%
  summarize(
    mean_length = mean(length_mm),
    sd_length = sd(length_mm),
    n = n(),
    se_length = sd_length / sqrt(n)
  )

print(group_summary)
```

```
# A tibble: 2 × 5
  side  mean_length sd_length     n se_length
  <chr>       <dbl>     <dbl> <int>     <dbl>
1 shady        17.6      2.51     8     0.886
2 sunny        16.2      2.64     8     0.934
```

Visualizing Group Differences

```
# Create a boxplot comparing the two sides
shady_sunny_plot
```



Practice Exercise 4: Effect Size

Practice Exercise 4: Effect size

We could also look at the difference in means… some cool code here

```
# Assuming your dataframe is called df
group_summary %>%
  summarize(difference = mean_length[side == "shady"] - mean_length[side == "sunny"])
```

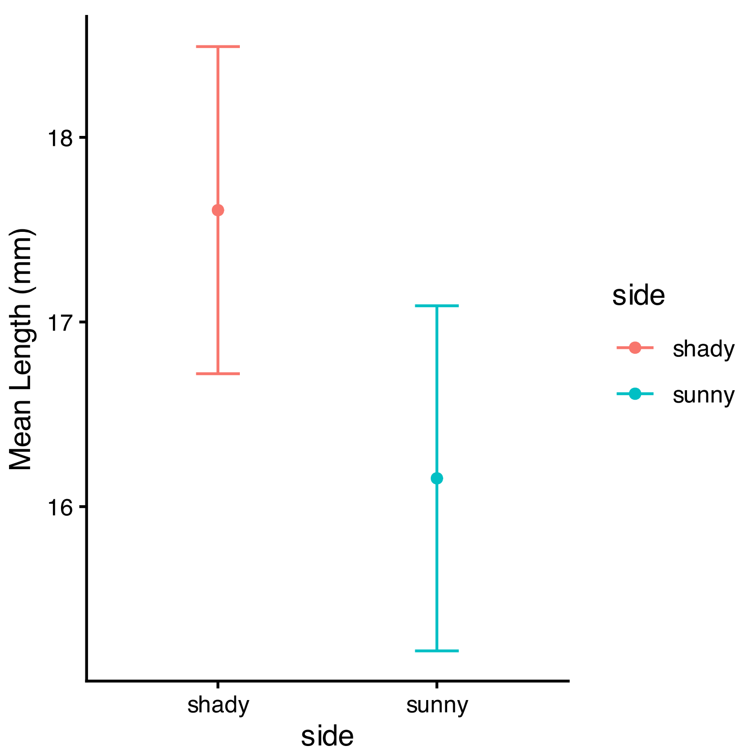
```
# A tibble: 1 × 1
  difference
       <dbl>
1       1.45
```

Practice Exercise 5: ggplot Summary Statistics

Practice Exercise 5: Using GGPLOT to get summary stats

GGplot also has code to make the mean and standard error plots we are interested in along with a lot of others

```
# Assuming your dataframe is called df
neelde_mean_se_plot <- ggplot(ps_df, aes(x = side, y = length_mm, color = side)) +
  stat_summary(fun = mean, geom = "point") +
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.2) +
  labs(
       x = "side",
       y = "Mean Length (mm)") +
  theme_classic()
neelde_mean_se_plot
```



Lecture 5: Testing Assumptions for Two-Sample T-Test

For a two-sample t-test, we need to check:

1. Normality within each group
2. Equal variances between groups (for standard t-test)
3. Independent observations

If assumptions are violated:

- Welch's t-test (unequal variances)
- Non-parametric alternatives (Mann-Whitney U test)

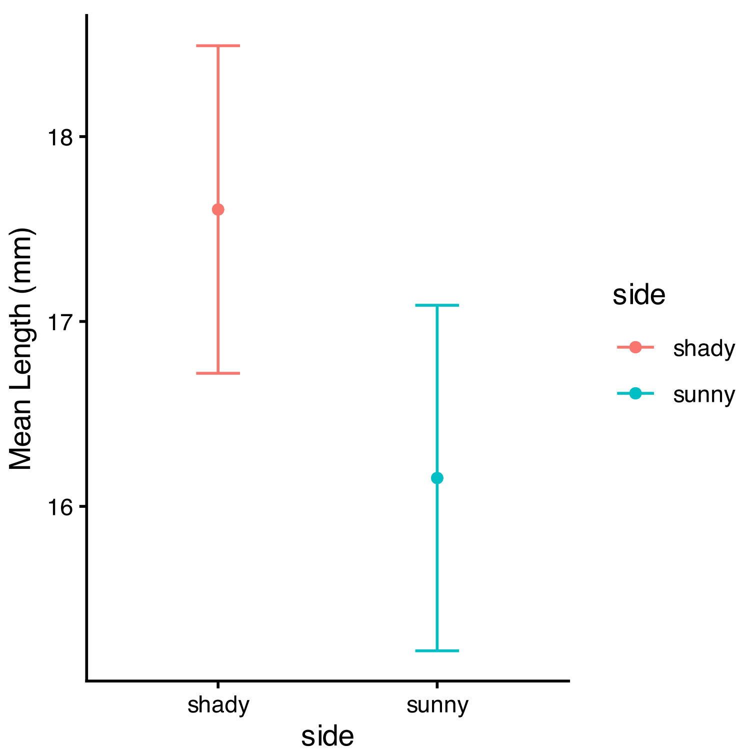
Practice Exercise 6: Creating Group Data

Practice Exercise 6: Test normality of windward pine needle lengths

qqplots

Note you need to test each groups separately…

```
# Assuming your dataframe is called df
neelde_mean_se_plot
```



Practice Exercise 7: Separate Group Data

Practice Exercise 7: Test normality of windward pine needle lengths

qqplots

Note you need to test each groups separately…

```
# how do you make separate dataframes to do this on?
# Separate data by groups
head(ps_shady_df)
```

```
# A tibble: 6 × 5
# Groups:   group, tree_no, tree_char [6]
  group                      tree_no tree_char side  length_mm
  <chr>                        <dbl> <chr>     <chr>     <dbl>
1 big_fat_fecund_female_fish       2 tree_2    shady      15.4
2 bill                             3 tree_3    shady      16.7
3 ciabatta                         5 tree_5    shady      19.1
4 fake_data                        8 tree_8    shady      17.4
5 five                             1 tree_1    shady      20.3
6 moose_walkin                     7 tree_7    shady      20.7
```

```
head(ps_sunny_df)
```

```
# A tibble: 6 × 5
# Groups:   group, tree_no, tree_char [6]
  group                      tree_no tree_char side  length_mm
  <chr>                        <dbl> <chr>     <chr>     <dbl>
1 big_fat_fecund_female_fish       2 tree_2    sunny      13.2
2 bill                             3 tree_3    sunny      16.0
3 ciabatta                         5 tree_5    sunny      17.7
4 fake_data                        8 tree_8    sunny      13.0
5 five                             1 tree_1    sunny      19.9
6 moose_walkin                     7 tree_7    sunny      18.4
```

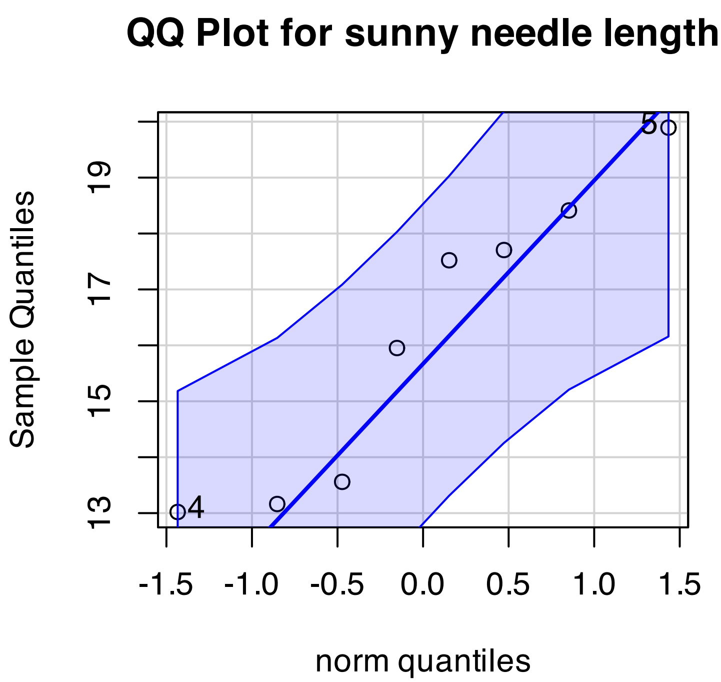
Practice Exercise 8: QQ Plot for Sunny Data

💡 Practice Exercise 8: Test normality of windward pine needle lengths

qqplots

Note you need to test each groups separately…

```
# QQ Plot for windward group
qqPlot(ps_sunny_df$length_mm,
       main = "QQ Plot for sunny needle length",
       ylab = "Sample Quantiles")
```




```
[1] 5 4
```

Practice Exercise 9: Shapiro-Wilk Test

Practice Exercise 9: Test normality of I8 lengths

Shapiro-Wilk test

Note you need to test each groups separately…

```
# Shapiro-Wilk test for windward group
shapiro.test(ps_sunny_df$length_mm)
```

```


	Shapiro-Wilk normality test

data:  ps_sunny_df$length_mm
W = 0.89994, p-value = 0.2886
```

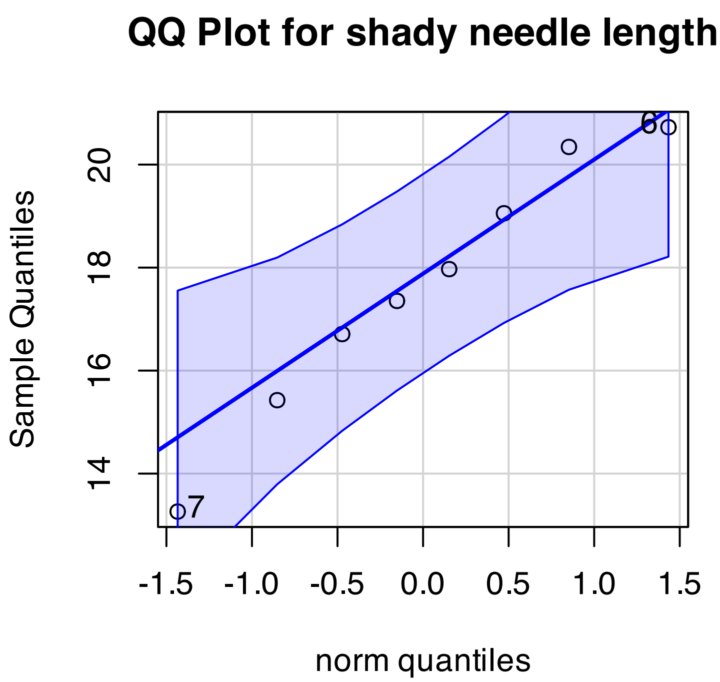
Practice Exercise 10: QQ Plot for Shady

Practice Exercise 10: Test normality of shady lengths

qqplots

Note you need to test each groups separately…

```
# You can also test the i3
# QQ Plot for leeward group
qqPlot(ps_shady_df$length_mm,
       main = "QQ Plot for shady needle length",
       ylab = "Sample Quantiles")
```




```
[1] 7 6
```

Practice Exercise 11: Shapiro-Wilk for Shady

Practice Exercise 11: Test normality of shady lengths

Shapiro-Wilk test

Note you need to test each groups separately…

```
# Shapiro-Wilk test for leeward group
 shapiro.test(ps_shady_df$length_mm)
```

```

	Shapiro-Wilk normality test

data:  ps_shady_df$length_mm
W = 0.96639, p-value = 0.8683
```

Practice Exercise 12: Combined Normality Test

Practice Exercise 12: Test Normality at one time

There are always a lot of ways to do this in R

```
# there are always two ways
# Test for normality using Shapiro-Wilk test for each wind group
# All in one pipeline using tidyverse approach
normality_results <- ps_df %>%
  group_by(side) %>%
  summarize(
    shapiro_stat = shapiro.test(length_mm)$statistic,
    shapiro_p_value = shapiro.test(length_mm)$p.value,
    normal_distribution = if_else(shapiro_p_value > 0.05, "Normal", "Non-normal")
  )

# Print the results
print(normality_results)
```

```
# A tibble: 2 × 4
  side  shapiro_stat shapiro_p_value normal_distribution
  <chr>        <dbl>           <dbl> <chr>
1 shady        0.966           0.868 Normal
2 sunny        0.900           0.289 Normal
```

Practice Exercise 13: Test Equal Variances

Practice Exercise 13: Test equal variances

Levenes test can be done on the original dataframe

```
# Method 1: Using car package's leveneTest
# This is often preferred as it's more robust to departures from normality
levene_result <- leveneTest(length_mm ~ side, data = ps_df)
print("Levene's Test for Homogeneity of Variance:")
```

```
[1] "Levene's Test for Homogeneity of Variance:"
```

```
print(levene_result)
```

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  1  0.2062 0.6567
      14
```

Lecture 5: Conducting the Two-Sample T-Test

Now we can compare the mean fish lengths between shady and sunny sides.

Ho: $\mu_1 = \mu_2$ (The needle lengths do not differ)

Ha: $\mu_1 \neq \mu_2$ (The mean needle lengths differ - direction is not specified)

Deciding between:

- Standard t-test (equal variances)
- Welch's t-test (unequal variances)

Note the Levenes Test should be NOT SIGNIFICANT - What is the null hypothesis

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  1  0.2062 0.6567
      14
```

Lecture 5: Running the Two-Sample T-Test

Now we can do a two sample TTEST

Calculate t-statistic manually (optional)

YOUR CODE HERE:

t = (mean1 - mean2) / sqrt((s1^2/n1) + (s2^2/n2))

Tip

```
# YOUR TASK: Conduct a two-sample t-test
# Use var.equal=TRUE for standard t-test or var.equal=FALSE for Welch's t-test

# Standard t-test (if variances are equal)
t_test_result <- t.test(length_mm ~ side, data = ps_df, var.equal = TRUE)
print("Standard two-sample t-test:")
```

```
[1] "Standard two-sample t-test:"
```

```
print(t_test_result)
```

```

	Two Sample t-test

data:  length_mm by side
t = 1.1279, df = 14, p-value = 0.2783
alternative hypothesis: true difference in means between group shady and group sunny is
not equal to 0
95 percent confidence interval:
 -1.309330  4.214005
sample estimates:
mean in group shady mean in group sunny
           17.60561            16.15328
```

```
# Welch's t-test (if variances are unequal)
# YOUR CODE HERE
```

Lecture 5: Running the Two-Sample T-Test

Now we can do a two sample TTEST

Calculate t-statistic manually (optional)

YOUR CODE HERE:

t = (mean1 - mean2) / sqrt((s1^2/n1) + (s2^2/n2))

Tip

```r
# YOUR TASK: Conduct a two-sample t-test
# Use var.equal=TRUE for standard t-test or var.equal=FALSE for Welch's t-test

# Standard t-test (if variances are equal)
t_test_result <- t.test(length_mm ~ side, data = ps_df, var.equal = FALSE)
print("Standard two-sample t-test:")
```

```
[1] "Standard two-sample t-test:"
```

```r
print(t_test_result)
```

```

	Welch Two Sample t-test

data:  length_mm by side
t = 1.1279, df = 13.96, p-value = 0.2784
alternative hypothesis: true difference in means between group shady and group sunny is
not equal to 0
95 percent confidence interval:
 -1.310069  4.214743
sample estimates:
mean in group shady mean in group sunny
           17.60561            16.15328
```

```r
# Welch's t-test (if variances are unequal)
# YOUR CODE HERE
```

Lecture 5: Difference between a Two-Sample T and Welch's T Test

Standard t-test (Student's t-test)

- **Assumes equal variances** between the two groups being compared
- Uses a **pooled variance estimate** that combines data from both group
- Has **higher statistical power** when the equal variance assumption is met
- **Degrees of freedom** = $n_1 + n_2 - 2$

Welch's t-test

- **Does not assume equal variances** between groups (also called the "unequal variances t-test")
- Uses **separate variance estimates** for each group
- More **robust** when group variances are different
- Uses a more complex **degrees of freedom calculation** (**Welch-Satterthwaite equation**) and decimal!!!
- **Degrees of freedom** are typically non-integer and usually smaller than the standard t-test

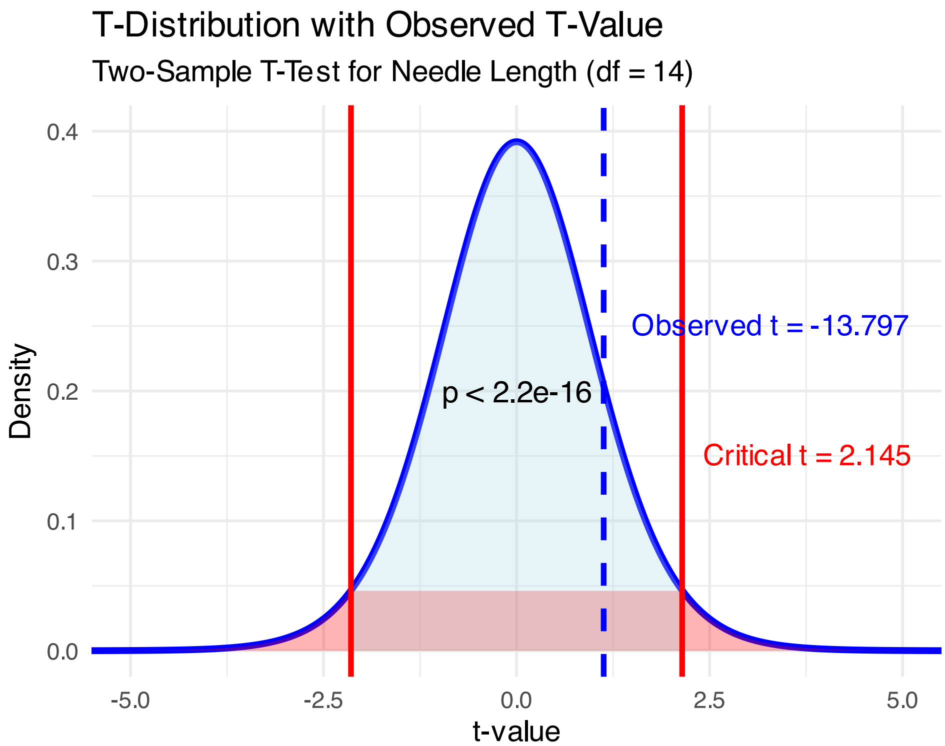
Lecture 5: Interpreting Two-Sample T-Test Results

Interpret the results of the two-sample t-test

What can we conclude about the needle lengths on sunny vs shady sides?

How to report this result in a scientific paper:

"A two-tailed, two-sample t-test at α=0.05 showed [a significant/no significant] difference in needle length between windward (M = …, SD = …) and leeward (M = …, SD = …) sides of pine trees, t(…) = …, p = …."



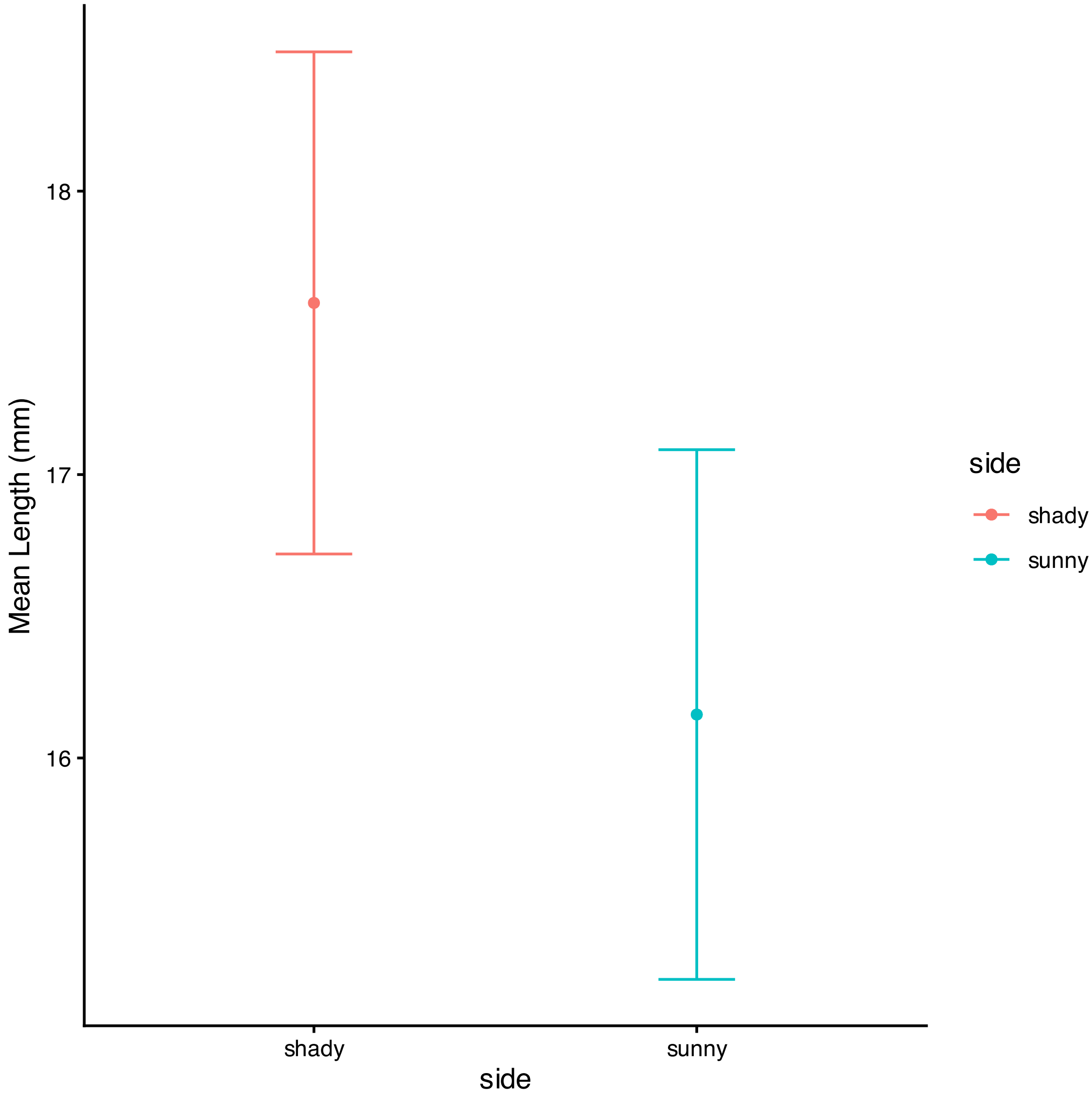
Lecture 5: Visualizing the Results

Interpret the results of the two-sample t-test

What can we conclude about the needle lengths on sunny vs shady sides?

How to report this result in a scientific paper:

"A two-tailed, two-sample t-test at α=0.05 showed [a significant/no significant] difference in needle length between windward (M = …, SD = …) and leeward (M = …, SD = …) sides of pine trees, t(…) = …, p = …."



Lecture 5: Now what does a paired T test tell us

Why do a paired T test

```
# YOUR TASK: Conduct a two-sample t-test
# Use var.equal=TRUE for standard t-test or var.equal=FALSE for Welch's t-test

ps_wide_df <- ps_df %>%
  pivot_wider(
    names_from = "side",
    values_from = length_mm
  )


# Standard t-test (if variances are equal)
paired_t_test_result <- t.test(ps_wide_df$sunny, ps_wide_df$shady, paired = TRUE)
print("Standard two-sample t-test:")
```

```
[1] "Standard two-sample t-test:"
```

```r
print(paired_t_test_result)
```

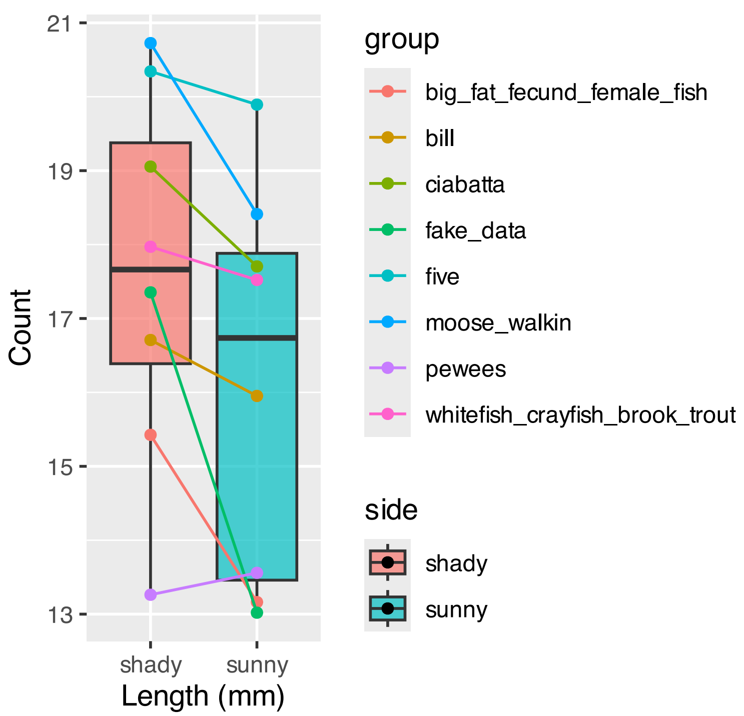
```

	Paired t-test

data:  ps_wide_df$sunny and ps_wide_df$shady
t = -2.7818, df = 7, p-value = 0.02723
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -2.6868652 -0.2178092
sample estimates:
mean difference
      -1.452337
```

```r
# Welch's t-test (if variances are unequal)
# YOUR CODE HERE
```

Lecture 5: What is going on??

Note that thee is a lot of variation within trees but the trend is the same

```r
ps_plot
```



Pine needles switched

Lecture 5: Assumptions of Parametric Tests

Common assumptions for t-tests:

1. Normality: Data comes from normally distributed populations
2. Equal variances (for two-sample tests)
3. Independence: Observations are independent

4. No outliers: Extreme values can influence results

What can we do if our data violates these assumptions?

Alternatives when assumptions are violated:

- Data transformation (log, square root, etc.)
- Non-parametric tests
- Robust statistical methods

Lecture 5: Summary and Conclusions

In this activity, we've:

1. Formulated hypotheses about pine needle length
2. Tested assumptions for parametric tests
3. Conducted one-sample and two-sample t-tests
4. Visualized data using appropriate methods
5. Learned how to interpret and report t-test results

Key takeaways:

- Always check assumptions before conducting tests
- Visualize your data to understand patterns
- Report results comprehensively
- Consider alternatives when assumptions are violated - non parametric tests…